Part A

Publication 1

Field	Details
Title	Predicting Hospital Length of Stay Using Machine Learning on a Large Open Health Dataset
Harvard Reference	Jain, R., Singh, M., Rao, A.R. & Garg, R., 2024. Predicting hospital length of stay using machine learning on a large open health dataset. BMC Health Services Research, 24(860).
Link	BMC Health Services Research Article
Modelling	They compared Random Forest, XGBoost, LightGBM, CatBoost, Linear Regression and Decision Trees, judging each one on standard regression metrics.
Preprocessing Techniques	The authors cleaned the data, dealt with missing values, encoded the categorical variables, split the data into training and test sets, and selected features.
Datasets Used	The New York State SPARCS (Statewide Planning and Research Cooperative System) dataset, which holds roughly 2.3 million de-identified patient records.
Key Takeaways	1. Machine learning can predict a patient's length of stay with good accuracy. 2. Tree-based models did better than the traditional regression approaches. 3. Knowing likely length of stay helps hospitals plan their resources and capacity.

Publication 2

Field	Details
Title	Predicting Hospital Length of Stay Using Machine Learning on a Large Open Health Dataset
Harvard Reference	Jain, R., Singh, M., Rao, A.R. & Garg, R., 2024. Predicting hospital length of stay using machine learning on a large open health dataset. BMC Health Services Research, 24(860).
Link	BMC Health Services Research Article
Modelling	They compared Random Forest, XGBoost, LightGBM, CatBoost, Linear Regression and Decision Trees, judging each one on standard regression metrics.
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Key Takeaways	1. Machine learning can predict a patient's length of stay with good accuracy. 2. Tree-based models did better than the traditional regression approaches. 3. Knowing likely length of stay helps hospitals plan their resources and capacity.

Publication 3

Field	Details
Title	Forecasting Hospital Room and Ward Occupancy Using Static and Dynamic Information Concurrently: Retrospective Single-Center Cohort Study
Harvard Reference	Cahan, E.M. et al., 2024. Forecasting hospital room and ward occupancy using static and dynamic information concurrently: retrospective single-center cohort study. JMIR Medical Informatics.
Link	JMIR Medical Informatics Article
Modelling	Occupancy forecasting models that bring together static patient-level features and dynamic operational information from the hospital.
Preprocessing Techniques	Admission sequences were prepared, features were pulled from patient and operational records, and the data was processed with time in mind.
Datasets Used	Retrospective ward occupancy and admission data from a hospital.
Key Takeaways	1. Patient-flow information makes it possible to predict occupancy. 2. Accuracy improves when patient data and operational data are combined. 3. Forecasts of this kind help with staffing and bed management ahead of time.

Public Dataset for Part B

Field	Details
Name	MIMIC-IV (Medical Information Mart for Intensive Care IV)
Harvard Reference	Johnson, A., Bulgarelli, L., Pollard, T., Gow, B., Moody, B., Horng, S., Celi, L.A. & Mark, R., 2023. *MIMIC-IV, a freely accessible electronic health record dataset*. Scientific Data, 10(1).
Link	https://physionet.org/content/mimiciv/
Number of Instances	The hospital module has 546,028 hospital admissions from 223,452 patients. The ICU module adds 94,458 ICU stays.
Number of Feature Columns	The admissions table records things like admission and discharge times, admission type, discharge location, insurance, marital status and race. Beyond that, the dataset is made up of many linked tables with hundreds of variables covering both clinical and operational information.
Number of Classes (if applicable)	Not applicable, as this would be a regression task (predicting Length of Stay).
Target Range (if applicable)	Length of Stay (LOS) can be worked out from the admission and discharge timestamps, which makes it a natural continuous target for regression.
Origin	MIMIC-IV is a de-identified electronic health record dataset collected at Beth Israel Deaconess Medical Center in Boston, USA. It covers hospital admissions, ICU stays, diagnoses, procedures, lab results and operational hospital data.